

INTRODUCTION TO SYSTEM ANALYSIS AND DESIGN

System Analysis and design (SAD) refers to one aspect of the process of creating or modifying an information system in order to meet the needs and goals of the given business system.

When a business decides that it has outgrown its current information system, it goes through the process of analysing and design as it attempts to remedy the problem.

Information system

It's a set of components that manages the data needed by a business system. It exists only to serve the business system of which it's a component.

It produces information required by managers to make decisions about the firm and must satisfy the information needs so that accurate and timely decisions can be made.

System Analysis

It is a process of collecting and interpreting facts, identifying the problems, and decomposition of a system into its components.

System analysis is conducted for the purpose of studying a system or its parts in order to identify its objectives.

It is a problem-solving technique that improves the system and ensures that all the components of the system work efficiently to accomplish their purpose.

Analysis specifies **what** the system should do.

This is all done by the system analyst.

Information Requirements

A detailed statement of the information needs that a new system must satisfy i.e. identifies who needs what information, when, where and how the information is needed.

Requirement analysis carefully defines the objectives of the new or modified system and develops a detailed description of the functions that the new system must perform.

Faulty requirements analysis is a leading cause of a system failure and high system development costs.

System Design

It is a process of planning a new business system or replacing an existing system by defining its components or modules to satisfy the specific requirements. Before planning, you need to understand the old system thoroughly and determine how computers can best be used in order to operate efficiently.

System Design focuses on **how to accomplish the objective of the system**.

This is done by the system designer.

NB: Analysis specifies **what** the system should do while design **how** to accomplish the objectives.

System analysis and design is the process of planning a business system or one to represent or compliment an existing system.

However before this planning can be done a thorough understanding of the old system is required in order to determine how best computers can be used to make operations more effective.

Role of the System Analyst

The system analyst systematically assesses how businesses function by examining the inputting and processing of data and the outputting of information with the intention of improving organizational processes.

Many improvements involve better support of business functions through the use of computerized information systems.

Primary roles of the Analyst are:

- a) *As a Consultant* - The analyst may be hired specifically to address information system issues within a business. Such hiring can be an advantage because outside consultant can bring with them a fresh perspective that other members of the organization do not possess.
- b) *As a Support Agent* - The analyst draws on professional expertise concerning hardware and software and their uses in the business. As the support expert he does not manage the project but serve as a resource for those who are.

- c) *As an Agent of Change* - An agent of change is the person who serves as a catalyst for change, develops a plan for change and works with others in facilitating that change. As an analyst you must recognise this fact and use it as a starting point for your analysis hence you must interact with users and management from the very beginning of your project. If change (i.e. improvements to the business that can be realised through information systems) seems warranted after analysis, next step is to develop a plan for change along with the people who must enact the change. Once a consensus is reached on the change that is to be made, you must constantly interact with those who are changing.

Change – technology, identifying problems, opportunities, objectives, people resist change.

- d) *As an Architect* - He designs and develop new system that are to be implemented within the organization.

Principle Duties of a System Analyst

- Carry out feasibility study on existing and proposed systems to determine the economic viability of computer processing within them.
- Prepare outline system specifications which show how the computer would operate and estimates the time required to develop and implement the proposed system.
- Assesses the resources required for, and total cost of preparing and installing the computer procedures and supporting manual systems develop in the outline specification.
- Checks operation of completed system and compares actual results against benefits claimed in the feasibility study.
- Monitors development progress through regular review meetings.
- Lays down standards and techniques to be observed in systems work and ensures conformity with them.
- Maintains current awareness of new developments in system technology and equipments that are relevant to his installation.
- Maintenance of close liason with the chief programmers during programming phase.
- Co-ordinate the implementation of new or modified system.

- Review the performance of the implemented systems and assessing the need for ammendments.

Characteristics of a Good Analyst

- *Exellent communication skills* - Acts as the translator between technical and staff and the non technical staff (users and managers) – people who frequently speak computer jargon and people who are baffled by it.
- *A diplomat and a good motivator* - He must try to elicit cooperation or enthusiasm from everyone involved in the project i.e. from team members to users.
- *Be able to work in project team both as a team member and as a team leader* - All team members must forget their ego i.e. put aside personal prejudices in order to help the team as a whole more effectively. As a team leader the analyst co-ordinates the efforts of team members as well as well resolving conflict and monitors the results of the team effort.
- *Must have well developed problem solving skills* - He should be able to identify symptoms, determine the cause of the symptoms and find a solution to the problem. All these require an organised approach to problem solving as well as a great deal of creativity.
- *Must be a business generalist* - A good analyst needs to know as much about business systems as about computer systems in order to be able to link the two e.g. a programmer might insist that for efficiency, he/she needs to use assembler language to code a particular module but the analyst having greater knowledge of how hard an assembler routine will be to maintain in the future, must convince the programmer that perhaps in this case, programming efficiency is not the primary concern.

SYSTEM THEORY

System

Is an orderly group of interdependent components linked together according to a plan to achieve a specific objective. The system breaks down whenever any of its components are removed hence a system is considered in its entirety.

The system does some useful tasks and should be active and efficient.

All systems have got a boundary within which they operate. Outside the boundary is the environment from which inputs are communicated and to which results are communicated. Each system has other systems that communicate with each other i.e. a sub system.

Importance of System Study

The study of a system is important for the following reasons:

- It provides a theoretical framework which allows performance of a system to be improved.
- It stresses that all systems must work in harmony in order to achieve the overall goals of a system.
- It recognises the fact that conflicts can arise in any system leading to sub optimization thus making the system not achieve its goals.
- It allows an individual to recognize that he/she is a subsystem within a big system.
- The design of a system must support the goals of the total system.
- Help identify the effect of the environment on a system. The external factors that affect an organisation include government, competitors, trade unions, creditors, shareholders etc and all have an interactive link with the organisation.

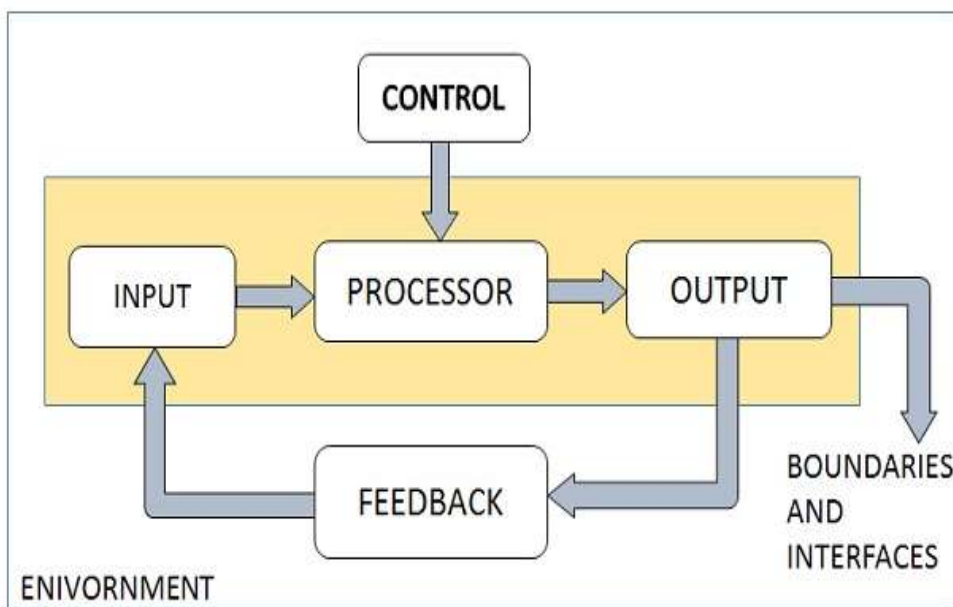
Characteristics of a System

A system is established to achieve some specific objectives. The main characteristics of a system include:

- it must be designed to achieve a predetermined objective.
- It must contain a set of interacting elements.

- It comprises of inputs, process and outputs.
- It operates within specified boundaries and interacts with other systems.
- It has some control and these controls help it not to operate beyond its boundaries.
- It must give priority to the objectives of the organization as a whole compared to the objectives of a subsystem.

Functions of a System



Outputs and Inputs

- The main aim of a system is to produce an output which is useful for its user.
- Inputs are the information that enters into the system for processing.
- Output is the outcome of processing.

Processor(s)

- The processor is the element of a system that involves the actual transformation of input into output.
- It is the operational component of a system. Processors may modify the input either totally or partially, depending on the output specification.
- As the output specifications change, so does the processing. In some cases, input is also modified to enable the processor for handling the transformation.

Control

- The control element guides the system.
- It is the decision-making subsystem that controls the pattern of activities governing input, processing, and output.
- The behaviour of a computer System is controlled by the Operating System and software. In order to keep system in balance, what and how much input is needed is determined by Output Specifications.

Feedback

- Feedback provides the control in a dynamic system.
- Positive feedback is routine in nature that encourages the performance of the system.
- Positive feedback encourages further output the way they are.
- Negative feedback is informational in nature that provides the controller with information for action.
- Negative feedback regulates the system when non standard output is achieved.
- It works on the principle of trying to reduce fluctuations around a set standard e.g. restricting credit limit of customers with outstanding debts.

Environment

- The environment is the “supersystem” within which an organization operates.
- It is the source of external elements that strike on the system.
- It determines how a system must function. For example, vendors and competitors of organization’s environment, may provide constraints that affect the actual performance of the business.

Boundaries and Interface

- A system should be defined by its boundaries. Boundaries are the limits that identify its components, processes, and interrelationship when it interfaces with another system.
- Each system has boundaries that determine its sphere of influence and control.
- The knowledge of the boundaries of a given system is crucial in determining the nature of its interface with other systems for successful design.

Description of System Principles

- *System Whole* - A system is an assembly of procedures , processes, methods, routines or techniques united by some regulated interaction to form an organised whole.
- *Function* - Each system is created to achieve a designed set of objectives for the organisation.
- *Inter-relatedness and interdependence of components* - All systems and sub systems are inter related and interdependent. When any element of a system is changed or eliminated, the rest of the system elements and subsystems are also impacted.
- *Sub systems* - Are the building blocks of the whole system. A ‘whole’ system must be broken down into smaller units for easier development. The subsystems communicate with each other.
- *Entropy* - It’s the amount of randomness or disorder that is present in the system. It brings about decay/confusion leading to obsolescence.

Types of Systems

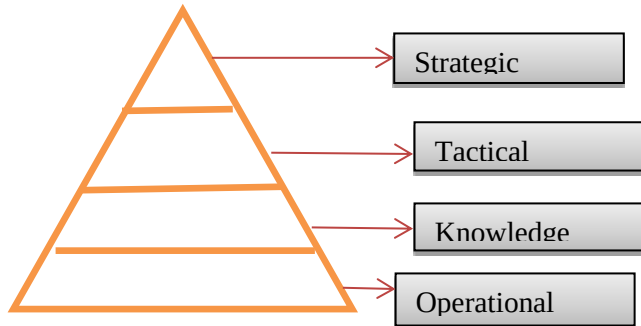
- i. *Open System* - It interacts and exchanges information with the environment receiving input and influence from the environment and its output is also influenced by the environment. The purpose of interaction is for information gathering or conducting business otherwise disorder will increase e.g. a business system.
- ii. *Closed System* - It does not interact with the environment. They don’t even exchange business transactions. Change in the environment does not affect it. They move to a state of internal equilibrium involving increased disorder.
- iii. *Deterministic System* - The outputs are known precisely from knowledge of inputs. They have predetermined rules of obtaining the outputs.
- iv. *Probabilistic System (Stochastic)*- Are those whose outputs can be predicted using probability and statistical methods. Have a lot of uncertainty and include business and economic systems.
- v. *Cybernetic Systems (Adaptive/Self Controlling)* - Adapt to changes in the environment and regulates their own behaviour and outputs using relevant inbuilt mechanism e.g human body – temperature changes.
- vi. *Open Loop Systems* - Output is determined and controlled by input and environment which behaves as an additional input. Are systems that act in no controlled manner i.e. no feedback and so no measure of performance against standards e.g. heater with a thermo start.
- vii. *Closed Loop Systems* - Are systems that act in controlled manner i.e. there is feedback. Feedback based on the output is fed back so as to make appropriate alterations to the input

e.g. stock control system - the actual stock levels are measured and compared with the plan and adjustments made to the stock replenishment order quantities to bring stock levels in line with the plan.

Information Systems

It's a set of interrelated components that collect or retrieve, process, store and distribute information to support decision making and control in an organisation.

Levels Of Organisational Information



Operational Level

- It deals with the day to day activities of an organisation.
- These are the basic activities that the organisation must perform in order to remain in business.
- Information is derived entirely from internal sources and is highly detailed e.g. a report of daily orders.
- Decisions associated with these activities cover a relatively narrow frame and are also delegated to the lowest possible level of the organisation where they can be made quickly and effectively.
- The decisions tend to be recurring and as a result the decision making process becomes routine and structured.
- A structured decision is one that is predictable and can be made following a well defined set of procedures.
- The decisions have an immediate but short term effect on the firm.
- Information must be easily available, current and accurate.

Knowledge Level

- Support knowledge and data workers in an organisation.
- They help interpret new knowledge into the business and help the organisation control the flow of paper work.
- These are fastest growing applications in business today (especially in office system).

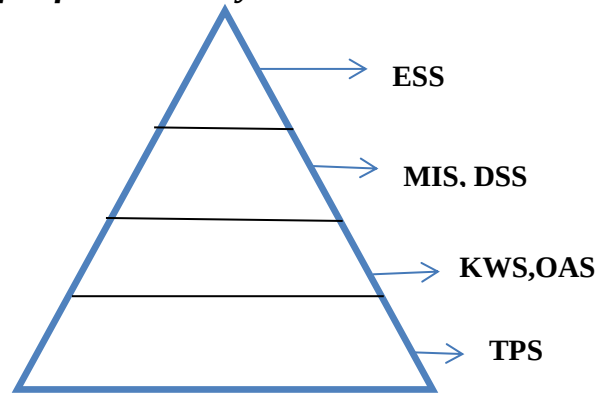
Tactical Level

- Middle managers make short term planning and control decisions about how resources may best be allocated to meet organisational objectives.
- The decisions range from forecasting future resource requirements to solving employee problems that threaten productivity.
- The decision making domain of middle managers can usefully be characterised as partly operational and partly strategic.
- The decisions are meant to implement the goals that the strategic management have set.

Strategic Planning

- It's concerned with setting long term goals and the decisions that provide guidelines on which the firm will run e.g. which strategy do we follow in competing against other firms.
- The strategic decisions are highly complex and unstructured.
- They have a high degree of uncertainty and such as demand the most experience and good sense of judgement on the part of managers.
- The decisions have a long term impact on the firm and it's difficult to reverse the impact of strategic decision.
- Information to make such decisions is derived from internal and external sources and is highly summarised.

Types Of Information Systems



Transaction Processing System (TPS)

- Serve the operational level of the organisation.
- It performs and records the daily routine transactions necessary to the conduct of the business e.g. sales order entry, employee record keeping.
- At operational level tasks, resources and goals are predefined and highly structured e.g. decision to grant credit to a customer is made by lower level supervisor according to predefined criteria. All that must be determined is whether the customer meets the criteria.

Knowledge Work Systems (KWS)

- They aid knowledge workers in the creation and interpretation of new knowledge in the organisation.
- They promote the creation of new knowledge and ensure that new knowledge and technical expertise are properly interpreted in the business.
- Knowledge workers are often members of recognised profession e.g. doctors, lawyers, scientists etc.

Office Automation Systems (OAS)

- They collect, process, store and transmit electronic messages, documents and other forms of communications among individuals, work groups and organisation.
- They aid data workers who tend to process rather than create information e.g. secretaries, accountants, managers etc.

- Typical OAS handle and manage documents (through word processing and desktop publishing), scheduling (through electronic calendars) and communication (through electronic mail, voice mail or video conferencing).

Management Information Systems (MIS)

- They serve the tactical/management level of the organisation that serve the functions of planning, controlling and decision making by providing routine summary and exception reports.
- Generally, they are dependent on underlying transaction processing systems for their data.
- The managers are interested in weekly, monthly and yearly results not day to day activities.

Characteristics of MIS

- They support structured and semi-structured decisions.
- Are generally reporting and control oriented. They are designed to report on existing operations and help to provide day to day control of operations.
- They rely on existing corporate data and data flows.
- Aid in decision making using past and present data.
- Have an internal rather than external orientation.
- Information requirements are known and stable.

Types of Reports Generated by MIS

- *Programmed/Scheduled/Periodic Reports* – are reports produced at pre-determined intervals e.g. weekly, fortnightly, monthly, quarterly, annually e.g. payroll, financial statements.
- *Exceptional Reports* – are reports which show out of the ordinary data like when production falls below specified level
- *On demand/Adhoc Reports* – are reports usually requested for by a manager when information is needed that focuses on a particular problem e.g. before giving a loan, a report on the history of the customer account may be requested for.
- *Event Initiated Report* – are reports that deal with a change in condition that require immediate action such as out of stock report.

- *Summary Reports* – are reports which show total and trends e.g. report showing total sales by office, product e.t.c.

Decision Support Systems (DSS)

- Are information system at the tactical level of an organisation that assist management decision making by combining data sophisticated analytical models and user friendly software to support semi-structured and unstructured decision making.
- Senior and middle management want a more interactive system that they could control and run themselves with as many changes in data and models as they need and with little interference from data processing professionals.

Characteristics Of A DSS

- i. Offers users flexibility, adaptability and a quick response.
- ii. Allow users to initiate and control the input and output.
- iii. Operate with little or no assistance from professional programmers.
- iv. Provide support for decisions and problem whose solutions cannot be specified in advance (unstructured decisions).

e.g.

- Senior management can use a financial DSS to forecast the availability of corporate funds for investments by departments.
- Middle managers within departments can use these estimates, the same systems and data to make decisions about allocating department fund to projects.
- Capital project managers within departments in turn can use these systems to begin their projects reporting to the system (and senior managers) on a regular basis about how much money has been spent.

Therefore DSS can be designed to facilitate group decision making.

Components Of DSS

- *Database* – collection of current or historical data from a number of applications or groups organised for easy access by a range of applications.
- *Model base* –collection of mathematical and analytical models that can easily be made accessible to the DSS users. A model is an abstract relationship that illustrates the components or relationships of a phenomenon.
- *DSS software system* –Permit easy interaction between the users of the system, the DSS database and model base. Provide a graphic easy to use flexible user interface that supports the dialog between the user and the DSS.
- *DSS User* – are users at tactical and strategic levels

Executive Support Systems (ESS)

- Are information systems at the strategic level of an organisation designed to address unstructured decision making through advanced graphics and communication.
- They filter, compress and track critical data emphasizing the reduction of time and effort required to obtain information useful to the executives.
- The systems have maintained the ability to drill down (ability to move from summary data down to lower and lower levels of details).
- It employs the most advanced graphics software (excel) and can deliver graphics and data from many sources immediately to senior executives.
- The software act as the heart of their analytical tool base from which they can create graphic comparisons of data by time, region, product, price range etc.
- ESS are not designed primarily to solve specific problems but instead they provide a generalised computing and telecommunications capacity that can be applied to changing array of problems.

How ESS Benefits Managers

- **Flexibility:** The systems put data and tools in the hands of executives without addressing specific problems or imposing solutions. They are free to shape the problems as they need using the system as an extension of their own thinking process.
- Ability to analyse, compare and highlight trends - look at data in less time with greater clarity and insight then paper based system can provide.

- Allow executives to monitor performance more successfully in their own areas of responsibility (timely data result in identified actions being taken)
- ESS can and do change the working of an organisations. Immediate access to so much data allows executives to better monitor activities of lower units reporting to them.